

PROGRAM 1: CALCULATE SIMPLE INTEREST

ALGORITHM:

```
Step 1: Start
Step 2: Read principle (P), rate (R), and time (T)
Step 3: Calculate SI = (P * R * T) / 100
Step 4: Print SI
Step 5: Stop
```

```
#include <stdio.h>

int main() {
    float p, r, t, si;

    printf("Enter principle amount: ");
    scanf("%f", &p);

    printf("Enter rate of interest: ");
    scanf("%f", &r);

    printf("Enter time period: ");
    scanf("%f", &t);

    si = (p * r * t) / 100;

    printf("Simple Interest = %f", si);

    return 0;
}
```

PROGRAM 2: CHECK EVEN OR ODD NUMBER

ALGORITHM:

```
Step 1: Start
Step 2: Input number N
Step 3: If N % 2 == 0 then print "Even"
Step 4: Else print "Odd"
Step 5: Stop
```

```
#include <stdio.h>

int main() {
    int n;

    printf("Enter a number: ");
    scanf("%d", &n);
```

```

    if (n % 2 == 0) {
        printf("%d is Even
", n);
    }
    else {
        printf("%d is Odd
", n);
    }

    return 0;
}

```

PROGRAM 3: FIND GREATER OF TWO NUMBERS

ALGORITHM:

```

Step 1: Start
Step 2: Input two numbers A and B
Step 3: If A > B print "A is greater"
Step 4: If B > A print "B is greater"
Step 5: If A == B print "Both are equal"
Step 6: Stop

```

```

#include <stdio.h>

int main() {
    int a, b;

    printf("Enter first number: ");
    scanf("%d", &a);

    printf("Enter second number: ");
    scanf("%d", &b);

    if (a > b) {
        printf("%d is greater
", a);
    }
    if (b > a) {
        printf("%d is greater
", b);
    }
    if (a == b) {
        printf("Both numbers are equal
");
    }

    return 0;
}

```

PROGRAM 4: BASIC CALCULATOR USING SWITCH CASE

ALGORITHM:

Step 1: Start
Step 2: Input operator choice (+, -, *, /)
Step 3: Input two numbers A and B
Step 4: Perform operation based on operator
Step 5: Print result
Step 6: Stop

```
#include <stdio.h>

int main() {
    char op;
    float a, b, res;

    printf("Enter operator (+, -, *, /): ");
    scanf(" %c", &op);

    printf("Enter two numbers: ");
    scanf("%f %f", &a, &b);

    switch(op) {
        case '+':
            res = a + b;
            printf("Result = %f", res);
            break;
        case '-':
            res = a - b;
            printf("Result = %f", res);
            break;
        case '*':
            res = a * b;
            printf("Result = %f", res);
            break;
        case '/':
            res = a / b;
            printf("Result = %f", res);
            break;
        default:
            printf("Invalid operator");
    }

    return 0;
}
```